



How Zephyr Projects Are Structured

- Standard layout: src/, prj.conf, CMakeLists.txt, and optional Devicetree overlays.
- prj.conf controls compile-time configuration through Kconfig.
- Devicetree overlays describe hardware, buses, and device connections.
- Build system uses CMake and west to generate the final executable.

```
/dts-v1/;  
  
/ {  
    soc {  
        i2c-bus-controller {  
            i2c-peripheral-1 {  
            };  
            i2c-peripheral-2 {  
            };  
            i2c-peripheral-3 {  
            };  
        };  
    };  
};
```

```
zephyr-esp32s3-course > apps > blinky > ⚙ prj.conf  
1  # esp32s3_demo/prj.conf  
2  CONFIG_GPIO=y  
3  CONFIG_LOG=y  
4  CONFIG_LOG_DEFAULT_LEVEL=3  
5  CONFIG_MAIN_STACK_SIZE=4096  
6  
7  # Comms  
8  
9  CONFIG_CONSOLE=y  
10 CONFIG_UART_CONSOLE=y  
11 CONFIG_SERIAL=y  
12 CONFIG_STDOUT_CONSOLE=y  
13  
14 # For debug
```

Key idea: Zephyr projects have clear app directories plus shared modules.